

Discrete Mathematics

□ Students will be able to develop logical thinking for solving computational problems.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3341 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3341.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Artificial Intelligence & Machine Learning
Course code	3341
Course title	Discrete Mathematics
Semester	3 Credits: 4
Course category	Program Core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

21 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Students will be able to develop logical thinking for solving computational problems.

Course outcomes

CO	Official outcome	Study level
CO1	16 Applying solve problems Applying	See official syllabus
CO2	Demonstrate sets, relations and functions 16 Solve counting problems by applying elementary	See official syllabus
CO3	16 Applying counting techniques	See official syllabus
CO4	Summarize the abstract algebraic systems 10 Understanding	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Apply the basic concepts of mathematical logic to solve problems	5	As specified
Module 2	Demonstrate sets, relations and functions	7	As specified
Module 3	Solve counting problems by applying elementary counting techniques	6	As specified
Module 4	Summarize the abstract algebraic systems	3	As specified

MODULE 1

Apply the basic concepts of mathematical logic to solve problems

This module is presented from the official syllabus scope for Discrete Mathematics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Apply the basic concepts of mathematical logic to solve problems
- Official duration: As specified
- Official topic entries captured: 5

- The full source excerpt is preserved below for coverage checking.

Describe the concept of propositional Logic 2 Understanding Solve problems using truth tables and logical

Describe the concept of propositional Logic 2 Understanding Solve problems using truth tables and logical is an official topic in Module 1 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the concept of propositional Logic 2 Understanding Solve problems using truth tables and logical using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the concept of propositional logic 2 understanding solve problems using truth tables and logical should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the concept of propositional Logic 2 Understanding Solve problems using truth tables and logical means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-
Describe the concept of propositional Logic 2
Understanding Solve problems using truth tables and logical
definition/meaning
definition/meaning steps, application
definition/meaning Technical terms English-
definition/meaning

3 Applying operations Prove logical equivalence, contingency,

3 Applying operations Prove logical equivalence, contingency, is an official topic in Module 1 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying operations Prove logical equivalence, contingency, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying operations prove logical equivalence, contingency, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying operations Prove logical equivalence, contingency, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-3 Applying operations Prove logical equivalence, contingency, definition/meaning steps, application Technical terms English-

4 Applying tautology and contradictions

4 Applying tautology and contradictions is an official topic in Module 1 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying tautology and contradictions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying tautology and contradictions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying tautology and contradictions means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-4 Applying tautology and contradictions

പ്രസ്താവനകളുടെ സാധാരണ definition/meaning പരിശോധിക്കുക. പ്രസ്താവനകളുടെ സാധാരണ, steps, application പരിശോധിക്കുക. Technical terms English-ൽ പരിശോധിക്കുക.

Describe the concept of predicate Logic

Describe the concept of predicate Logic is an official topic in Module 1 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the concept of predicate Logic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the concept of predicate logic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the concept of predicate Logic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Describe the concept of predicate Logic
definition/meaning . steps, application . Technical terms English-
.

Apply inference rules to solve problems 4 Applying Mathematical logic: Propositional logic, Connectives and truth table, Implication, Tautology, Contradiction, Contingency, Equivalence - inverse, converse, contrapositive, Duality principle, Disjunction and Conjunction, Predicate Logic, Well formed formula, Quantifiers-universal & existential, Natural deduction, Rules of Inference, Methods of proofs, Resolution principle.

Apply inference rules to solve problems 4 Applying Mathematical logic: Propositional logic, Connectives and truth table, Implication, Tautology, Contradiction, Contingency, Equivalence - inverse, converse, contrapositive, Duality principle, Disjunction and Conjunction, Predicate Logic, Well formed formula, Quantifiers-universal & existential, Natural deduction, Rules of Inference, Methods of proofs, Resolution principle. is an official topic in Module 1 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Apply inference rules to solve problems 4 Applying Mathematical logic: Propositional logic, Connectives and truth table, Implication, Tautology, Contradiction, Contingency, Equivalence - inverse, converse, contrapositive, Duality principle, Disjunction and Conjunction, Predicate Logic, Well formed formula, Quantifiers-universal & existential, Natural deduction, Rules of Inference, Methods of proofs, Resolution principle. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, apply inference rules to solve problems 4 applying mathematical logic: propositional logic, connectives and truth table, implication, tautology, contradiction, contingency, equivalence - inverse, converse, contrapositive, duality principle, disjunction and conjunction, predicate logic, well formed formula, quantifiers-universal & existential, natural deduction, rules of inference, methods of

proofs, resolution principle. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Apply inference rules to solve problems 4 Applying Mathematical logic: Propositional logic, Connectives and truth table, Implication, Tautology, Contradiction, Contingency, Equivalence - inverse, converse, contrapositive, Duality principle, Disjunction and Conjunction, Predicate Logic, Well formed formula, Quantifiers-universal & existential, Natural deduction, Rules of Inference, Methods of proofs, Resolution principle. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-എപ് Apply inference rules to solve problems 4 Applying Mathematical logic: Propositional logic, Connectives and truth table, Implication, Tautology, Contradiction, Contingency, Equivalence - inverse, converse, contrapositive, Duality principle, Disjunction and Conjunction, Predicate Logic, Well formed formula, Quantifiers-universal & existential, Natural deduction, Rules of Inference, Methods of proofs, Resolution principle. ഏകദേശം ഏകദേശം definition/meaning ഏകദേശം. ഏകദേശം ഏകദേശം ഏകദേശം, steps, application ഏകദേശം ഏകദേശം ഏകദേശം. Technical terms English-എ ഏകദേശം ഏകദേശം.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Apply the basic concepts of mathematical logic to solve problems

M1.01	Describe the concept of propositional Logic	2
Understanding		
	Solve problems using truth tables and logical	
M1.02		3
Applying		
	operations	
	Prove logical equivalence, contingency,	
M1.03		4
Applying		
	tautology and contradictions	
M1.04	Describe the concept of predicate Logic	3
Understanding		
M1.05	Apply inference rules to solve problems	4
Applying		
Contents:		
Mathematical logic:		
Propositional logic, Connectives and truth table, Implication, Tautology,		
Contradiction,		
Contingency, Equivalence - inverse, converse, contrapositive, Duality principle,		
Disjunction		
and Conjunction,		
Predicate Logic, Well formed formula, Quantifiers-universal & existential,		
Natural		
deduction, Rules of Inference, Methods of proofs, Resolution principle.		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Demonstrate sets, relations and functions

This module is presented from the official syllabus scope for Discrete Mathematics.
Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Demonstrate sets, relations and functions
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

Outline the basic principles of set

Outline the basic principles of set is an official topic in Module 2 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the basic principles of set using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the basic principles of set should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the basic principles of set means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
Outline the basic principles of set
definition/meaning .
steps, application . Technical terms English-
.

Demonstrate the operations on sets

Demonstrate the operations on sets is an official topic in Module 2 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate the operations on sets using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate the operations on sets should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate the operations on sets means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Demonstrate the operations on sets

പ്രദർശിക്കുക definition/meaning പ്രദർശിക്കുക. പ്രദർശിക്കുക പ്രദർശിക്കുക, steps, application പ്രദർശിക്കുക പ്രദർശിക്കുക. Technical terms English- പ്രദർശിക്കുക പ്രദർശിക്കുക.

Apply inclusion exclusion principle

Apply inclusion exclusion principle is an official topic in Module 2 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Apply inclusion exclusion principle using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, apply inclusion exclusion principle should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Apply inclusion exclusion principle means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്ന പ്രിൻസിപ്പിൾ

പ്രവർത്തനങ്ങൾക്ക് നിർവ്വചന/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചന പ്രവർത്തനങ്ങൾ, steps, application നിർവ്വചന പ്രവർത്തനങ്ങൾ നിർവ്വചിക്കുക. Technical terms English-ൽ നിർവ്വചിക്കുക.

Demonstrate the concept of relations

Demonstrate the concept of relations is an official topic in Module 2 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate the concept of relations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate the concept of relations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate the concept of relations means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
Demonstrate the concept of relations
definition/meaning .
steps, application . Technical terms English-
.

Identify the equivalence relations 2 Applying Illustrate the equivalence classes from

Identify the equivalence relations 2 Applying Illustrate the equivalence classes from is an official topic in Module 2 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify the equivalence relations 2 Applying Illustrate the equivalence classes from using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify the equivalence relations 2 applying illustrate the equivalence classes from should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify the equivalence relations 2 Applying Illustrate the equivalence classes from means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$ Identify the equivalence relations 2 Applying
Illustrate the equivalence classes from $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$ definition/meaning
 $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$. $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$, steps, application $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$
 $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$. Technical terms English- $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$

3 Applying equivalence relation

3 Applying equivalence relation is an official topic in Module 2 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying equivalence relation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying equivalence relation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying equivalence relation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Applying equivalence relation $\sim$
 $\sim$ definition/meaning $\sim$. $\sim$ $\sim$ $\sim$, steps,
application $\sim$ $\sim$ $\sim$. Technical terms English- $\sim$ $\sim$
 $\sim$.

Demonstrate the concept of functions 3 Applying Set theory:-Set representation, Types of Sets, Set operations, Algebra of sets, Inclusion-exclusion principle Relations: Representation - matrix -graph, types of relations, properties, equivalence relations, equivalence class Functions: Mappings, Injection and Surjections, Composition of functions, Inverse functions

Demonstrate the concept of functions 3 Applying Set theory:-Set representation, Types of Sets, Set operations, Algebra of sets, Inclusion- exclusion principle Relations: Representation - matrix -graph, types of relations, properties, equivalence relations, equivalence class Functions: Mappings, Injection and Surjections, Composition of functions, Inverse functions is an official topic in Module 2 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate the concept of functions 3 Applying Set theory:-Set representation, Types of Sets, Set operations, Algebra of sets, Inclusion- exclusion principle Relations: Representation - matrix -graph, types of relations, properties, equivalence relations, equivalence class Functions: Mappings, Injection and Surjections, Composition of functions, Inverse functions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate the concept of functions 3 applying set theory:-set representation, types of sets, set operations, algebra of sets, inclusion- exclusion principle relations: representation - matrix -graph, types of relations, properties, equivalence relations, equivalence class functions: mappings, injection and surjections, composition of functions, inverse functions should be discussed with attention to safe

operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate the concept of functions 3 Applying Set theory:-Set representation, Types of Sets, Set operations, Algebra of sets, Inclusion- exclusion principle Relations: Representation - matrix -graph, types of relations, properties, equivalence relations, equivalence class Functions: Mappings, Injection and Surjections, Composition of functions, Inverse functions means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Demonstrate the concept of functions 3 Applying Set theory:-Set representation, Types of Sets, Set operations, Algebra of sets, Inclusion- exclusion principle Relations: Representation - matrix -graph, types of relations, properties, equivalence relations, equivalence class Functions: Mappings, Injection and Surjections, Composition of functions, Inverse functions
 definition/meaning .
 , steps, application . Technical terms English-
 .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Demonstrate sets, relations and functions

M2.01	Outline the basic principles of set	2
Understanding		
M2.02	Demonstrate the operations on sets	2
Applying		
M2.03	Apply inclusion exclusion principle	2
Applying		
M2.04	Demonstrate the concept of relations	2
Understanding		
M2.05	Identify the equivalence relations	2
Applying		
	Illustrate the equivalence classes from	
M2.06		3
Applying		
	equivalence relation	
M2.07	Demonstrate the concept of functions	3
Applying		
	Series Test – I	1
Contents:		
Set theory:-	Set representation, Types of Sets, Set operations, Algebra of sets, Inclusion-exclusion principle	
Relations:	Representation - matrix -graph, types of relations, properties, equivalence relations, equivalence class	
Functions:	Mappings, Injection and Surjections, Composition of functions, Inverse	

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Solve counting problems by applying elementary counting techniques

This module is presented from the official syllabus scope for Discrete Mathematics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Solve counting problems by applying elementary counting techniques
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

Applying techniques using the product and sum rules

Applying techniques using the product and sum rules is an official topic in Module 3 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Applying techniques using the product and sum rules using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, applying techniques using the product and sum rules should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Applying techniques using the product and sum rules means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- $\square$ Applying techniques using the product and sum rules $\square$ $\square$ definition/meaning $\square$. $\square$ $\square$ $\square$ $\square$ $\square$, steps, application $\square$ $\square$ $\square$. Technical terms English- $\square$ $\square$ $\square$.

Applying permutations & combinations

Applying permutations & combinations is an official topic in Module 3 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Applying permutations & combinations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, applying permutations & combinations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Applying permutations & combinations means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Applying permutations & combinations

日本語から英語へ訳す definition/meaning 日本語から英語へ訳す。 日本語から英語へ訳す, steps, application 日本語から英語へ訳す 日本語から英語へ訳す。 Technical terms English-日本語 日本語から英語へ訳す。

Applying pigeon-hole principle 2

Applying pigeon-hole principle 2 is an official topic in Module 3 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Applying pigeon-hole principle 2 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, applying pigeon-hole principle 2 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Applying pigeon-hole principle 2 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എപ്പലൈംഗ് പൈൻ-ഹോൾ പ്രിൻസിപ്പിൾ 2 എപ്പലൈംഗ് പൈൻ-ഹോൾ പ്രിൻസിപ്പിൾ
എപ്പലൈംഗ് definition/meaning എപ്പലൈംഗ് പ്രിൻസിപ്പിൾ. എപ്പലൈംഗ് എപ്പലൈംഗ് എപ്പലൈംഗ്, steps,
application എപ്പലൈംഗ് എപ്പലൈംഗ് എപ്പലൈംഗ്. Technical terms English-എപ്പലൈംഗ്
എപ്പലൈംഗ്.

Explain Binomial theorem Understanding Make use of recurrence relations to solve 3

Explain Binomial theorem Understanding Make use of recurrence relations to solve 3 is an official topic in Module 3 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Binomial theorem Understanding Make use of recurrence relations to solve 3 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain binomial theorem understanding make use of recurrence relations to solve 3 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Binomial theorem Understanding Make use of recurrence relations to solve 3 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- $\square$ Explain Binomial theorem Understanding Make use of recurrence relations to solve 3 $\square$ $\square$ definition/meaning $\square$. $\square$ $\square$ $\square$, steps, application $\square$ $\square$. Technical terms English- $\square$ $\square$ $\square$.

Applying problems 2

Applying problems 2 is an official topic in Module 3 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Applying problems 2 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, applying problems 2 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Applying problems 2 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എപ്ലൈയിംഗ് പ്രോബ്ലംസ് 2 ഏകദേശം/അർത്ഥം നിർവ്വചനം/മനസ്സിലാക്കൽ. ഏകദേശം നിർവ്വചനം, steps, application ഏകദേശം നിർവ്വചനം നിർവ്വചനം. Technical terms English-എപ്ലൈയിംഗ് പ്രോബ്ലംസ് നിർവ്വചനം.

Illustrate generating functions Applying Basic counting principle, Rule of Sum, Rule of Product Permutations, Combinations. Pigeon hole Principle , Binomial Theorem (without proof). Recurrence relation, Generating functions

Illustrate generating functions Applying Basic counting principle, Rule of Sum, Rule of Product Permutations, Combinations. Pigeon hole Principle , Binomial Theorem (without proof). Recurrence relation, Generating functions is an official topic in Module 3 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate generating functions Applying Basic counting principle, Rule of Sum, Rule of Product Permutations, Combinations. Pigeon hole Principle , Binomial Theorem (without proof). Recurrence relation, Generating functions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate generating functions applying basic counting principle, rule of sum, rule of product permutations, combinations. pigeon hole principle , binomial theorem (without proof). recurrence relation, generating functions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate generating functions Applying Basic counting principle, Rule of Sum, Rule of Product Permutations, Combinations. Pigeon hole Principle , Binomial Theorem (without proof). Recurrence relation, Generating functions means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഇംഗ്ലീഷ് Illustrate generating functions Applying Basic counting principle, Rule of Sum, Rule of Product Permutations, Combinations. Pigeon hole Principle , Binomial Theorem (without proof). Recurrence relation, Generating functions ഇംഗ്ലീഷ് definition/meaning ഇംഗ്ലീഷ്. ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്, steps, application ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്. Technical terms English-ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Solve counting problems by applying elementary counting techniques

		3	
M3.01	Solve counting problems by applying the techniques using the product and sum rules		Applying
M3.02	Solve counting problems by applying permutations & combinations	3	Applying
M3.03	Solve counting problems by applying the pigeon-hole principle	3	Applying
M3.04	Explain Binomial theorem	2	
Understanding			
M3.05	Make use of recurrence relations to solve problems	3	Applying
M3.06	Illustrate generating functions	2	Applying
Contents: Basic counting principle, Rule of Sum, Rule of Product Permutations, Combinations. Pigeon hole Principle , Binomial Theorem (without proof). Recurrence relation, Generating functions			

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Summarize the abstract algebraic systems

This module is presented from the official syllabus scope for Discrete Mathematics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize the abstract algebraic systems
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Understanding isomorphism Describe the abstract algebraic systems - 5

Understanding isomorphism Describe the abstract algebraic systems - 5 is an official topic in Module 4 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding isomorphism Describe the abstract algebraic systems - 5 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding isomorphism describe the abstract algebraic systems - 5 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding isomorphism Describe the abstract algebraic systems - 5 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-Understanding isomorphism Describe the abstract algebraic systems - 5 definition/meaning application, steps, application. Technical terms English-

Groups, subgroups, Semigroups, Monoids & Understanding abelian groups
Outline the concept of lattice by definition, 3

Groups, subgroups, Semigroups, Monoids & Understanding abelian groups
Outline the concept of lattice by definition, 3 is an official topic in Module 4 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Groups, subgroups, Semigroups, Monoids & Understanding abelian groups Outline the concept of lattice by definition, 3 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, groups, subgroups, semigroups, monoids & understanding abelian groups outline the concept of lattice by definition, 3 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Groups, subgroups, Semigroups, Monoids & Understanding abelian groups Outline the concept of lattice by definition, 3 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- ഗ്രൂപ്പുകൾ, Subgroups, Semigroups, Monoids & Understanding abelian groups Outline the concept of lattice by definition, 3 ഉദാഹരണങ്ങൾ ഉപയോഗിച്ച് definition/meaning വ്യക്തമാക്കുക. ഗ്രൂപ്പിന്റെ ഉപഗ്രൂപ്പ്, steps, application ഉദാഹരണം ഉപയോഗിച്ച് വ്യക്തമാക്കുക. Technical terms English-ൽ ഉപയോഗിക്കുക.

Understanding properties Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup, subgroup, monoid, abelian group (simple theorems) Lattice -Definition, properties, Types of lattice Text / Reference: T/R Book Title / Author T1 Discrete and combinatorial mathematics(An applied introduction),Ralph P Grimaldi, B V Ramana, 5th edition, Pearson T2 Kenneth H.Rosen,Discrete Mathematics and its application with combinatorics and graph theory,seventh edition,MGH,2011 T3 “Discrete Mathematics with applications”,Thomas Koshy,Academic press,2005 Online Resources: Sl. No. Website Link 1 <https://archive.nptel.ac.in/courses/111/106/111106086/>

Understanding properties Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup, subgroup, monoid, abelian group (simple theorems) Lattice -Definition, properties, Types of lattice Text / Reference: T/R Book Title / Author T1 Discrete and combinatorial mathematics(An applied introduction),Ralph P Grimaldi, B V Ramana, 5th edition, Pearson T2 Kenneth H.Rosen,Discrete Mathematics and its application with combinatorics and graph theory,seventh edition,MGH,2011 T3 “Discrete Mathematics with applications”,Thomas Koshy,Academic press,2005 Online Resources: Sl. No. Website Link 1 <https://archive.nptel.ac.in/courses/111/106/111106086/> is an official topic in Module 4 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding properties Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup, subgroup, monoid, abelian group (simple theorems) Lattice -Definition, properties, Types of lattice Text / Reference: T/R Book Title / Author T1 Discrete and combinatorial mathematics(An applied introduction),Ralph P Grimaldi, B V Ramana, 5th edition, Pearson T2 Kenneth H.Rosen,Discrete Mathematics and its application with combinatorics and graph theory,seventh edition,MGH,2011 T3 “Discrete Mathematics with applications”,Thomas Koshy,Academic press,2005 Online Resources: Sl. No. Website Link 1 <https://archive.nptel.ac.in/courses/111/106/111106086/> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding properties algebraic system - properties, homomorphism and isomorphism, group, semigroup, subgroup, monoid, abelian group (simple theorems) lattice -definition, properties, types of lattice text / reference: t/r book title / author t1 discrete and combinatorial mathematics(an applied introduction),ralph p grimaldi, b v ramana, 5th edition, pearson t2 kenneth h.rosen,discrete mathematics and its application with combinatorics and graph theory,seventh edition,mgh,2011 t3 “discrete mathematics with applications”,thomas koshy,academic press,2005 online resources: sl. no. website link 1 <https://archive.nptel.ac.in/courses/111/106/111106086/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding properties Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup, subgroup, monoid, abelian group (simple theorems) Lattice -Definition, properties, Types of lattice Text / Reference: T/R Book Title / Author T1 Discrete and combinatorial mathematics(An applied introduction),Ralph P Grimaldi, B V Ramana, 5th edition, Pearson T2 Kenneth H.Rosen,Discrete Mathematics and its application with combinatorics and graph theory,seventh edition,MGH,2011 T3 “Discrete Mathematics with applications”,Thomas Koshy,Academic press,2005 Online Resources: Sl. No. Website Link 1 https://archive.nptel.ac.in/courses/111/106/111106086/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Understanding properties Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup, subgroup, monoid, abelian group (simple theorems) Lattice -Definition, properties, Types of lattice Text / Reference: T/R Book Title / Author T1 Discrete and combinatorial mathematics(An applied introduction),Ralph P Grimaldi, B V Ramana, 5th edition, Pearson T2 Kenneth H.Rosen,Discrete Mathematics and its application with

combinatorics and graph theory, seventh edition, MGH, 2011 T3 “Discrete Mathematics with applications”, Thomas Koshy, Academic press, 2005 Online Resources: Sl. No. Website Link 1 <https://archive.nptel.ac.in/courses/111/106/111106086/> ഏകീകൃതതയുടെ നിർവ്വചനം/അർത്ഥം ഏകീകൃതതയുടെ നിർവ്വചനം, steps, application ഏകീകൃതതയുടെ നിർവ്വചനം. Technical terms English-ഏകീകൃതതയുടെ നിർവ്വചനം.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Summarize the abstract algebraic systems		
Differentiate homomorphism and		2
M4.01		
Understanding	isomorphism	
	Describe the abstract algebraic systems -	5
M4.02	Groups, subgroups, Semigroups, Monoids &	
Understanding	abelian groups	
	Outline the concept of lattice by definition,	3
M4.03		
Understanding	properties	
	Series test II	1
Contents:		
Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup, subgroup, monoid, abelian group (simple theorems)		
Lattice -Definition, properties, Types of lattice		
Text / Reference:		
T/R	Book Title / Author	
T1	Discrete and combinatorial mathematics(An applied introduction), Ralph	
P		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Describe the concept of propositional Logic 2 Understanding Solve problems using truth tables and logical. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the concept of propositional Logic 2 Understanding Solve problems using truth tables and logical*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Applying operations Prove logical equivalence, contingency,. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying operations Prove logical equivalence, contingency,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Applying tautology and contradictions. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying tautology and contradictions*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Describe the concept of predicate Logic. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the concept of predicate Logic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain Outline the basic principles of set. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the basic principles of set*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Demonstrate the operations on sets. (3 marks)

Answer: Begin with the standard definition or identity of *Demonstrate the operations on sets*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Apply inclusion exclusion principle. (3 marks)

Answer: Begin with the standard definition or identity of *Apply inclusion exclusion principle*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Demonstrate the concept of relations. (3 marks)

Answer: Begin with the standard definition or identity of *Demonstrate the concept of relations*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain Applying techniques using the product and sum rules. (3 marks)

Answer: Begin with the standard definition or identity of *Applying techniques using the product and sum rules*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Applying permutations & combinations. (3 marks)

Answer: Begin with the standard definition or identity of *Applying permutations & combinations*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Applying pigeon-hole principle 2. (3 marks)

Answer: Begin with the standard definition or identity of *Applying pigeon-hole principle*
2. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain Binomial theorem Understanding Make use of recurrence relations to solve 3. (3 marks)

Answer: Begin with the standard definition or identity of *Explain Binomial theorem Understanding Make use of recurrence relations to solve 3*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain Understanding isomorphism Describe the abstract algebraic systems - 5. (3 marks)

Answer: Begin with the standard definition or identity of *Understanding isomorphism Describe the abstract algebraic systems - 5*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Groups, subgroups, Semigroups, Monoids & Understanding abelian groups Outline the concept of lattice by definition, 3. (3 marks)

Answer: Begin with the standard definition or identity of *Groups, subgroups, Semigroups, Monoids & Understanding abelian groups* Outline the concept of lattice by definition, 3. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Understanding properties Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup, subgroup, monoid, abelian group (simple theorems) Lattice -Definition, properties, Types of lattice Text / Reference: T/R Book Title / Author T1 Discrete and combinatorial mathematics(An applied introduction),Ralph P Grimaldi, B V Ramana, 5th edition, Pearson T2 Kenneth H.Rosen,Discrete Mathematics and its application with combinatorics and graph theory,seventh edition,MGH,2011 T3 “Discrete Mathematics with applications”,Thomas Koshy,Academic press,2005 Online Resources: Sl. No. Website Link 1 <https://archive.nptel.ac.in/courses/111/106/111106086/>. (3 marks)

Answer: Begin with the standard definition or identity of *Understanding properties Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup, subgroup, monoid, abelian group (simple theorems) Lattice -Definition, properties, Types of lattice* Text / Reference: T/R Book Title / Author T1 Discrete and combinatorial mathematics(An applied introduction),Ralph P Grimaldi, B V Ramana, 5th edition, Pearson T2 Kenneth H.Rosen,Discrete Mathematics and its application with combinatorics and graph theory,seventh edition,MGH,2011 T3 “Discrete Mathematics with applications”,Thomas Koshy,Academic press,2005 Online Resources: Sl. No. Website Link 1 <https://archive.nptel.ac.in/courses/111/106/111106086/>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Describe the concept of propositional Logic 2 Understanding Solve problems using truth tables and logical.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Outline the basic principles of set.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Applying techniques using the product and sum rules.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Understanding isomorphism Describe the abstract algebraic systems - 5.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No. Website Link
- <https://archive.nptel.ac.in/courses/111/106/111106086/>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 3341. Verify institutional updates against the current official curriculum.